

the risk aversion of  $\gamma = 3$ , the indifference curve has the steepest slope. If we increase risk (standard deviation), this investor needs to be compensated the most so the indifference curve shoots sharply upward as risk increases. In contrast, the investor with  $\gamma = 0.5$  is relatively risk tolerant: this investor does not need to be compensated as much with high expected returns when risk is increased. Thus, her indifference curves are relatively flat. It is important to note that an investor has one risk aversion parameter and a family of indifference curves (as drawn in Figure 2.5). Figure 2.6 compares indifference curves for different investors.

### 3.2. CERTAINTY EQUIVALENT

The utility certainty equivalent is the sure amount of wealth, or the risk-free return, that makes the investor feel the same as holding a risky asset position.<sup>16</sup> The certainty equivalent is the compensation the investor requires that would make him relinquish a risky asset.

For mean-variance utility, the indifference curve provides a convenient way to compute the certainty equivalent. An investor has the same utility along all combinations of risk and return on an indifference curve, by definition. The intersection at the  $y$ -axis of an indifference curve represents an asset with a positive return and no volatility—a risk-free asset. Thus, we can trace the indifference curve going through a risky asset position to the  $y$ -axis intercept to find the certainty equivalent.

Figure 2.7 plots the mean and standard deviation of the volatility strategy, which is marked with the square, and the indifference curve that goes through the volatility strategy for a  $\gamma = 3$  investor. The certainty equivalent of the volatility strategy is 6.45%. This means that if the investor could not invest \$1 in the volatility strategy, the investor would need to be compensated 6.45 cents on that \$1; the volatility strategy is equivalent to a risk-free return of 6.45%. The certainty equivalent is also called *willingness to pay*. An asset owner would be willing to pay 6.45 cents per dollar of wealth (risk-free) to be able to have the same utility as investing in the volatility strategy (which is risky).

Mean-variance utility ignores the large negative skew of the volatility strategy in computing this certainty equivalent. The mean and standard deviation of the volatility strategy are almost the same as the stock market, so this would be approximately the same certainty equivalent of the S&P 500 as well.

When we compute the certainty equivalent with mean-variance utility, we make use of the fact that the mean-variance utility function in equation (2.4) has economic meaning stated in terms of its level. In equation (2.4), the utility level itself is the certainty equivalent value. That is, mean-variance utility exhibits

<sup>16</sup>Formally the certainty equivalent is the amount  $C$  satisfying  $U(C) = E[U(X)]$  for the risky lottery  $X$ .